

Here are study flashcards based on the content of **Sessions 4 & 5: Processes**:

Flashcard 1: What is a Process?

Q: What is a process in an operating system?

A: A process is an instance of a running program that includes the program code and its current activity.

Flashcard 2: Preemptive vs. Non-preemptive Processes

Q: What is the difference between preemptive and non-preemptive processes?

A:

- **Preemptive:** Can be forcibly interrupted by the OS to allow another process to execute.
 - **Non-preemptive:** Runs until completion without interruption.
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Flashcard 3: Process vs. Thread

Q: How do processes differ from threads?

A:

- **Processes:** Independent execution units with their own memory and resources.
 - **Threads:** Lightweight sub-units within a process that share memory and resources.
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Flashcard 4: Process Life Cycle States

Q: What are the five states in a process life cycle?

A: New, Ready, Running, Blocked, Terminated.

Flashcard 5: Schedulers

Q: What are the three types of schedulers?

A:

1. **Short-term Scheduler:** Selects processes for CPU execution.
 2. **Medium-term Scheduler:** Temporarily removes processes to manage system load.
 3. **Long-term Scheduler:** Decides which processes to admit into the system.
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Flashcard 6: First Come, First Serve (FCFS)

Q: What is the main disadvantage of the FCFS scheduling algorithm?

A: Long waiting times for processes that arrive later.

Flashcard 7: Shortest Job First (SJF)

Q: What is the advantage of SJF scheduling?

A: Minimizes the average waiting time.

Flashcard 8: Round Robin (RR)

Q: How does the Round Robin scheduling algorithm work?

A: Each process gets a fixed time slice (quantum) for execution, cycling through all processes.

Flashcard 9: Multilevel Queue Scheduling

Q: What is multilevel queue scheduling?

A: Divides processes into multiple queues based on characteristics like priority or type.

Flashcard 10: Belady's Anomaly

Q: What is Belady's Anomaly?

A: Increasing the number of page frames can sometimes increase the number of page faults (e.g., in FIFO page replacement).

Flashcard 11: fork() System Call

Q: What does the `fork()` system call do?

A: Creates a new child process by duplicating the parent process.

Flashcard 12: waitpid() System Call

Q: What is the purpose of `waitpid()`?

A: Makes the parent process wait for a specific child process to terminate.

Flashcard 13: exec() System Call

Q: What happens when the `exec()` system call is used?

A: It replaces the current process image with a new program.

Flashcard 14: Orphan vs. Zombie Processes

Q: What is the difference between an orphan and a zombie process?

A:

- **Orphan Process:** Parent has terminated, adopted by the `init` process.

- **Zombie Process:** Terminated process still listed in the process table as the parent hasn't read its exit status.
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Let me know if you'd like additional details or adjustments!